SESSION 8 REFERENCE GUIDE: EXPLORATORY DATA ANALYSIS (EDA) WORKFLOW

This guide serves as a comprehensive reference for the complete EDA pipeline, consolidating data auditing, cleaning, relational merging, feature engineering, and multi-dimensional aggregation techniques.

1. The 5-Stage EDA Pipeline Reference

```python

import pandas as pd
import numpy as np

# STAGE 1: DATA AUDIT
df.shape                  # Check row/column geometry
df.info()                 # Check metadata, data types, and non-null counts
df.isnull().sum()         # Audit missing values per column
df.duplicated().sum()     # Audit exact duplicate rows

# STAGE 2: SANITIZATION
df = df.drop_duplicates().reset_index(drop=True)      # Deduplicate
df["text_col"] = df["text_col"].str.strip().str.title()# Standardize
strings
df["date_col"] = pd.to_datetime(df["date_col"])        # Cast datetime
median_val = df["numeric_col"].median()
df["numeric_col"] = df["numeric_col"].fillna(median_val)# Median Imputation

# STAGE 3: RELATIONAL MERGERS
df_merged = pd.merge(df_txns, df_meta, on="shared_key", how="left")

# STAGE 4: FEATURE ENGINEERING & MULTI-DIMENSIONAL EDA
df_merged["net_profit"] = df_merged["revenue"] -
df_merged["operating_cost"]
df_merged["margin_pct"] = (df_merged["net_profit"] / df_merged["revenue"])
* 100

summary_table = df_merged.groupby(["region", "category"]).agg(
    total_revenue=("revenue", "sum"),
    total_profit=("net_profit", "sum"),
    avg_margin=("margin_pct", "mean"),
    order_count=("txn_id", "count")
).sort_values(by="total_profit", ascending=True)

# STAGE 5: SYNTHESIS & REPORTING
# Convert summary_table findings into business implications and
recommendations.
```

2. EDA Checklist for Analysts

- [ ] Have you verified row counts before and after deduplication?
- [ ] Are date columns explicitly formatted as `datetime64[ns]`?
- [ ] Have categorical text columns been stripped of whitespace and standardized?
- [ ] Have missing values been imputed transparently using median baselines?
- [ ] Did you perform a Left Join to preserve all transaction records?
- [ ] Did you engineer net margin/profit metrics rather than relying solely on top-line sales?
- [ ] Are your aggregated tables sorted to highlight bottom/top performers immediately?